Splitting the $L_{1/2}$&TV Problem into Three Subproblems

Compound-regularized linear inverse problem [1], with echo data $\mathbf{y}_f$, observation operator $\mathbf{\Phi}$, scene $\mathbf{x}$, total variation $\mathrm{TV}(\cdot)$ [2] and penalty weights $\lambda_1, \lambda_2 > 0$. Introducing auxiliary variables [3] and then the Lagrangian with multipliers $\xi_1, \xi_2 > 0$ (kept fixed):

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{x}} \left\|\mathbf{y}_f - \mathbf{\Phi}\mathbf{x}\right\|_2^2 + \lambda_1 \|\mathbf{x}\|_{1/2} + \lambda_2 \,\mathrm{TV}(\mathbf{x}) \tag{1}$$

$$= \arg\min_{\mathbf{x},\mathbf{z}_1,\mathbf{z}_2} \left\|\mathbf{y}_f - \mathbf{\Phi}\mathbf{x}\right\|_2^2 + \lambda_1 \|\mathbf{z}_1\|_{1/2} + \lambda_2 \,\mathrm{TV}(\mathbf{z}_2) \quad \text{s.t.} \quad \|\mathbf{x} - \mathbf{z}_1\|_2^2 = 0, \quad \|\mathbf{x} - \mathbf{z}_2\|_2^2 = 0 \tag{2}$$

(introduce auxiliary variables $\mathbf{z}_1, \mathbf{z}_2$: $\|\mathbf{x} - \mathbf{z}\|_2^2 = 0 \iff \mathbf{z} = \mathbf{x}$)

$$\approx \arg\min_{\mathbf{x},\mathbf{z}_1,\mathbf{z}_2} \underbrace{\left\|\mathbf{y}_f - \mathbf{\Phi}\mathbf{x}\right\|_2^2 + \lambda_1 \|\mathbf{z}_1\|_{1/2} + \lambda_2 \,\mathrm{TV}(\mathbf{z}_2) + \xi_1 \|\mathbf{x} - \mathbf{z}_1\|_2^2 + \xi_2 \|\mathbf{x} - \mathbf{z}_2\|_2^2}_{\Gamma(\mathbf{x},\mathbf{z}_1,\mathbf{z}_2)} \tag{3}$$

(Lagrangian: move the constraints into the objective)

The minimization over $(\mathbf{x}, \mathbf{z}_1, \mathbf{z}_2)$ returns its $\mathbf{x}$-component. With ξ_1, ξ_2 fixed, $\xi\|\mathbf{x} - \mathbf{z}\|_2^2$ penalizes a copy that drifts away from $\mathbf{x}$ [3], so (3) approximates (2).

Alternating minimization of Γ (one variable at a time, the others at their latest values; terms without that variable are constants) splits (3) into three subproblems:

$$\mathbf{x}^{(k+1)} = \arg\min_{\mathbf{x}} \; \Gamma\big(\mathbf{x}, \mathbf{z}_1^{(k)}, \mathbf{z}_2^{(k)}\big) = \arg\min_{\mathbf{x}} \; \left\|\mathbf{y}_f - \mathbf{\Phi}\mathbf{x}\right\|_2^2 + \xi_1 \left\|\mathbf{x} - \mathbf{z}_1^{(k)}\right\|_2^2 + \xi_2 \left\|\mathbf{x} - \mathbf{z}_2^{(k)}\right\|_2^2, \tag{4}$$

$$\mathbf{z}_1^{(k+1)} = \arg\min_{\mathbf{z}_1} \; \Gamma\big(\mathbf{x}^{(k+1)}, \mathbf{z}_1, \mathbf{z}_2^{(k)}\big) = \arg\min_{\mathbf{z}_1} \; \xi_1 \left\|\mathbf{z}_1 - \mathbf{x}^{(k+1)}\right\|_2^2 + \lambda_1 \|\mathbf{z}_1\|_{1/2}, \tag{5}$$

$$\mathbf{z}_2^{(k+1)} = \arg\min_{\mathbf{z}_2} \; \Gamma\big(\mathbf{x}^{(k+1)}, \mathbf{z}_1^{(k+1)}, \mathbf{z}_2\big) = \arg\min_{\mathbf{z}_2} \; \xi_2 \left\|\mathbf{z}_2 - \mathbf{x}^{(k+1)}\right\|_2^2 + \lambda_2 \,\mathrm{TV}(\mathbf{z}_2). \tag{6}$$

- (4): quadratic in $\mathbf{x}$, solved by a linear system.
- (5): $L_{1/2}$ denoising of $\mathbf{x}^{(k+1)}$, solved entrywise by half thresholding [4].
- (6): TV denoising of $\mathbf{x}^{(k+1)}$, solved by Chambolle's projection algorithm [5].

References

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